

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMAL PROPERTIES OF MATTER

LECTURE-02 **PHYSICS**

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Topics To Be Covered

- 1 Revision ✓
- 2 Variation of density with temperature
- 3 Thermal Stress ✓
- 4 Abnormal behaviour of water ✓ (chemistry)

Recap from 1st Lecture

Heat

heat is the form of energy which produce the sensation hotness or coolness



Temperature

The degree of hotness or coolness

Kelvins, degree, Celsius, F

$$K = ^\circ C + 273$$

$$F = \frac{9}{5} ^\circ C + 32$$



Thermal Expansion



- ✓ Solid
- ✓ liquid
- ✓ Gases

✓ Coefficient of linear expansion α

$$l = l_0 + l_0 \alpha \Delta T$$

✓ Coefficient of superficial expansion β

$$A = A_0 + A_0 \beta \Delta T$$

✓ Coefficient of volume expansion γ

$$V = V_0 + V_0 \gamma \Delta T$$

Variation of density with temp.

lecture . no. ②

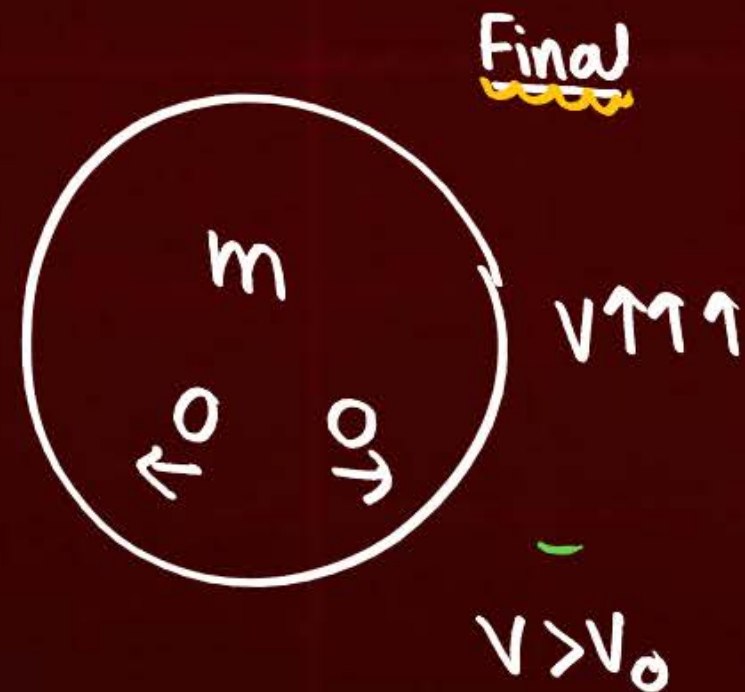


①



$$\downarrow \downarrow \rho = \frac{m}{V \uparrow}$$

- when temperature of a Body is increase the Body start expanding hence volume increase then density decrease



Temp ↑ Volume ↑↑ Density ↓↓↓

MCQ 2023

2019 Neet



Maths



m

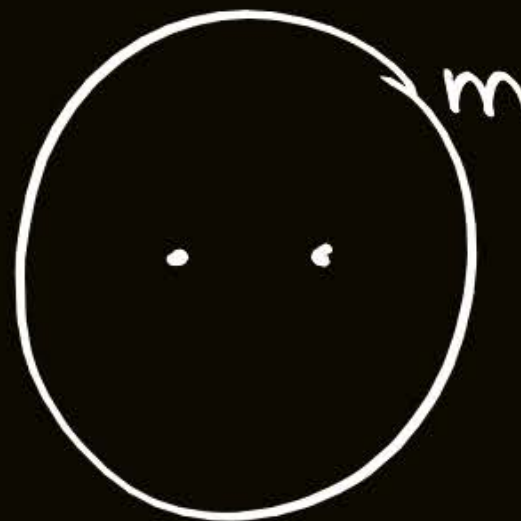
ρ_0 (initial Density)

V_0 (initial Volume)

↑↑↑

Temperature ↑↑↑

$$\rho_0 = \frac{m}{V_0}$$



m

ρ (Final Density) ↓↓↓

V (Final volume) ↑↑

$$\rho = \frac{m}{V}$$

$$\text{Density} = \frac{\text{mass}}{\text{Volume}}$$

$$\rho = \frac{m}{V}$$

$$V = V_0 + V_0 \gamma \Delta T$$

②

$$\rho = \frac{m}{V_0 + V_0 \gamma \Delta T}$$

$$\rho = \frac{m}{V_0(1 + \gamma \Delta T)}$$

$$\rho = \left(\frac{m}{V_0} \right) \frac{\rho_0}{(1 + \gamma \Delta T)}$$

$$\rho = \frac{\rho_0}{(1 + \gamma \Delta T)}$$

$$P = \frac{P_0}{(1+Y\Delta T)^1}$$

$$P = P_0 (1+Y\Delta T)^{-1}$$

$$P = P_0 (1 + -1Y\Delta T)$$

$$P = P_0 (1 - Y\Delta T)$$

Binomial

$$(1+x)^n = 1+nx$$

$$P = P_0 (1 - Y\Delta T)$$

$$P = P_0 - P_0 Y\Delta T$$

New

old

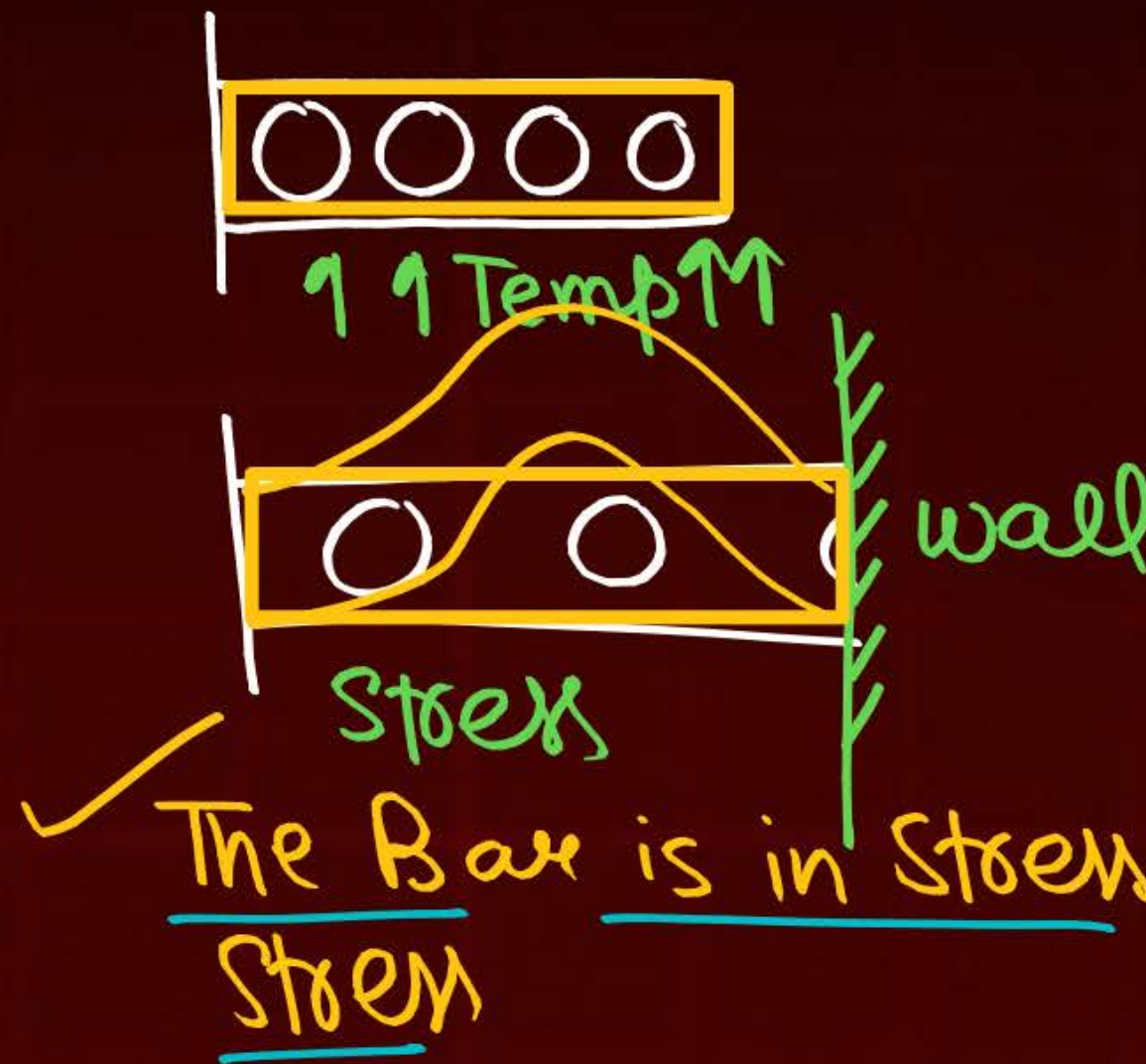
$$8 = 10 - 2$$

③ ✓

Thermal Stress

• when you heat the Bar (Iron) it expands, Volume increases ↑↑↑↑↑
But when you apply the Resistance Wall

→ it tries to increase its length But its length do not increase.



Formula of thermal stress

Neet-2021

- Young modulus = $\frac{\text{Stress}}{\text{Strain}}$

$$Y = \frac{\text{Thermal stress}}{\text{Strain}}$$

$$Y \times \text{Strain} = \text{Thermal stress}$$

$$\text{Thermal stress} = Y \times \text{Strain}$$

$$\text{Thermal stress} = Y \times \frac{\Delta l}{l_0}$$

$$\text{Strain} = \frac{\Delta l}{l_0} \checkmark$$

⑤

$$TS = Y \times \left(\frac{\Delta l}{l_0} \right) - \text{Jee-2023} \checkmark$$

$$TS = Y \alpha \Delta T$$

Thermal Stress

JKRGSE 2023

→ 2021 Neet

$Y \Rightarrow$ Young modulus

$\alpha \Rightarrow$ coefficient of linear expansion

$\Delta T \rightarrow$ change in temp

$$l = l_0 + l_0 \alpha \Delta T$$

$$\underline{l - l_0 = l_0 \alpha \Delta T}$$

$$\Delta l = \underline{l_0} \alpha \Delta T$$

$$\left(\frac{\Delta l}{l_0} \right) = \alpha \Delta T$$

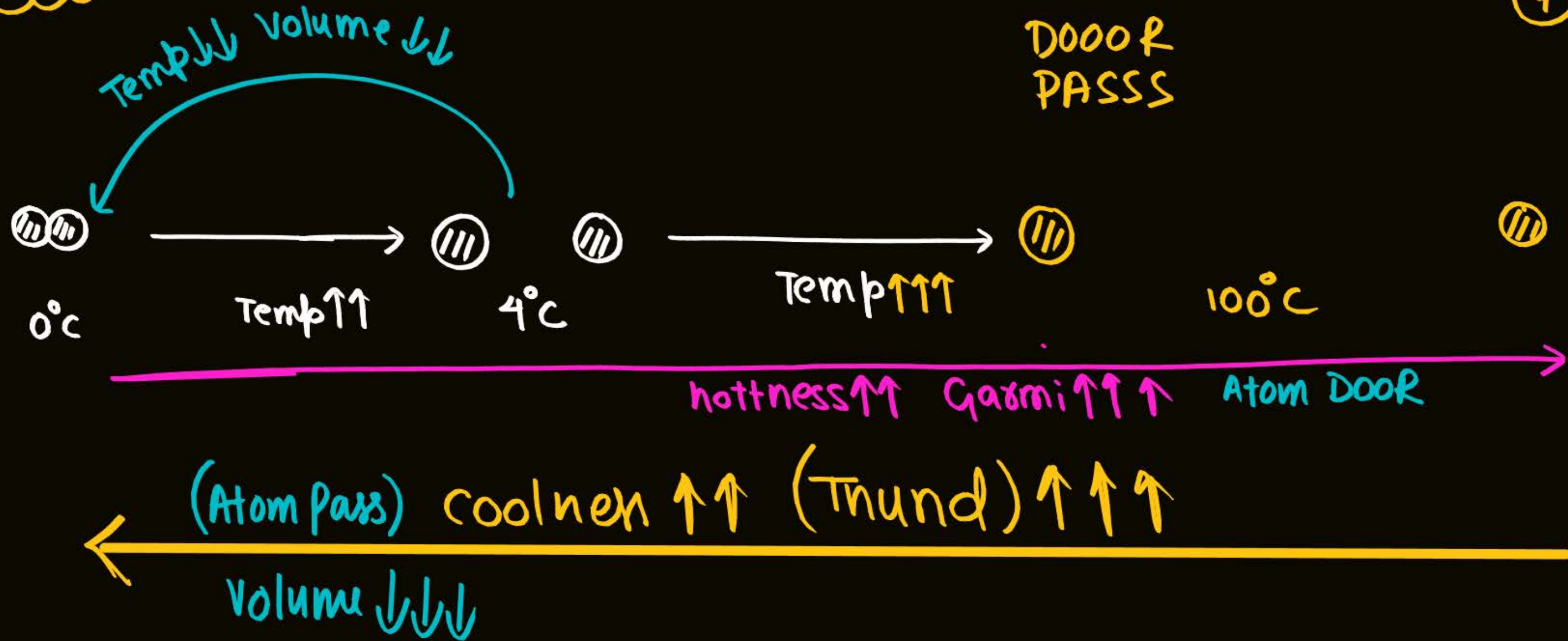
$$\text{Strain} = \alpha \Delta T$$

last class

⑥

Note. General

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Abnormal behaviour of water



- when you decrease the temperature from 4°C to 0°C

Generally volume decrease ↓↓

But in water volume increase

This is the Abnormal
Behaviour of water



water (ice) $0^{\circ}\text{C} \rightarrow f \downarrow \downarrow \downarrow$

(10)

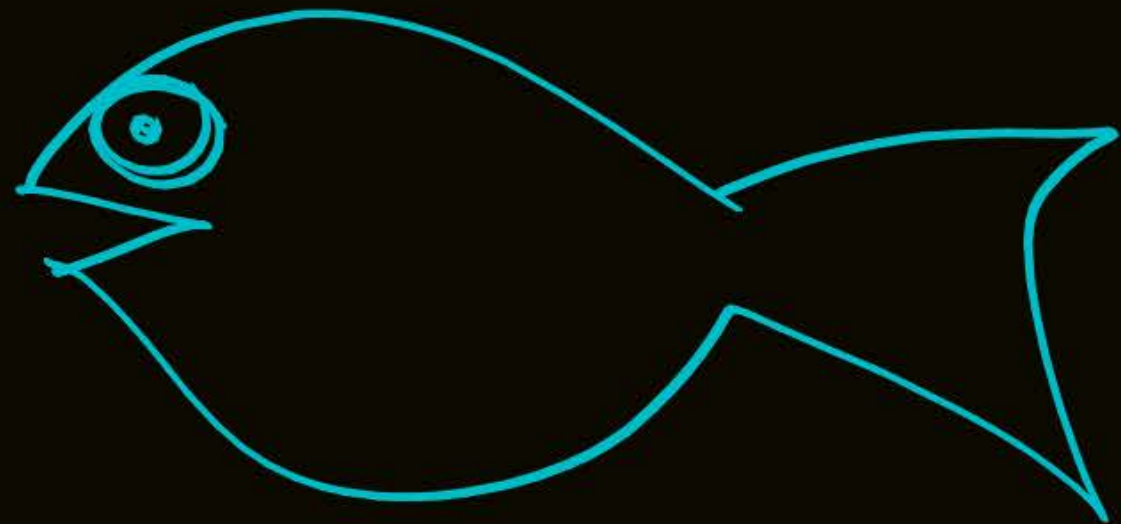
water (liquid) $4^{\circ}\text{C} \rightarrow f_{\text{max}} \uparrow \uparrow$

Dal-lake

ice $f \downarrow \downarrow$ 0°C

4°C

water
 $f \uparrow \uparrow$
(max)



• water (ice)
0°C



$v \uparrow \uparrow f \downarrow$



water (liquid)
4°C



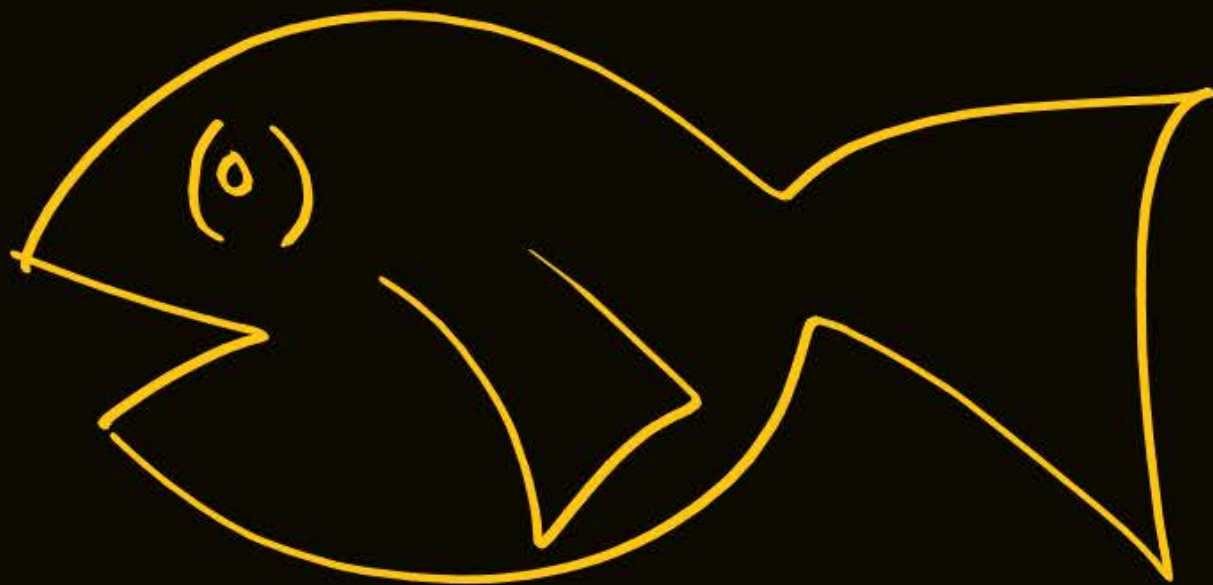
$v \downarrow \downarrow f_{\max} \uparrow \uparrow$

②

0°C ice (water)

$f \downarrow \downarrow \downarrow$

float



water
4°C
 $f_{\max}$

Q The water has _____ volume at 4°C ?

12

(a) ~~Minimum~~

(b) Maximum

Q The water has _____ Density at 4°C ?

13

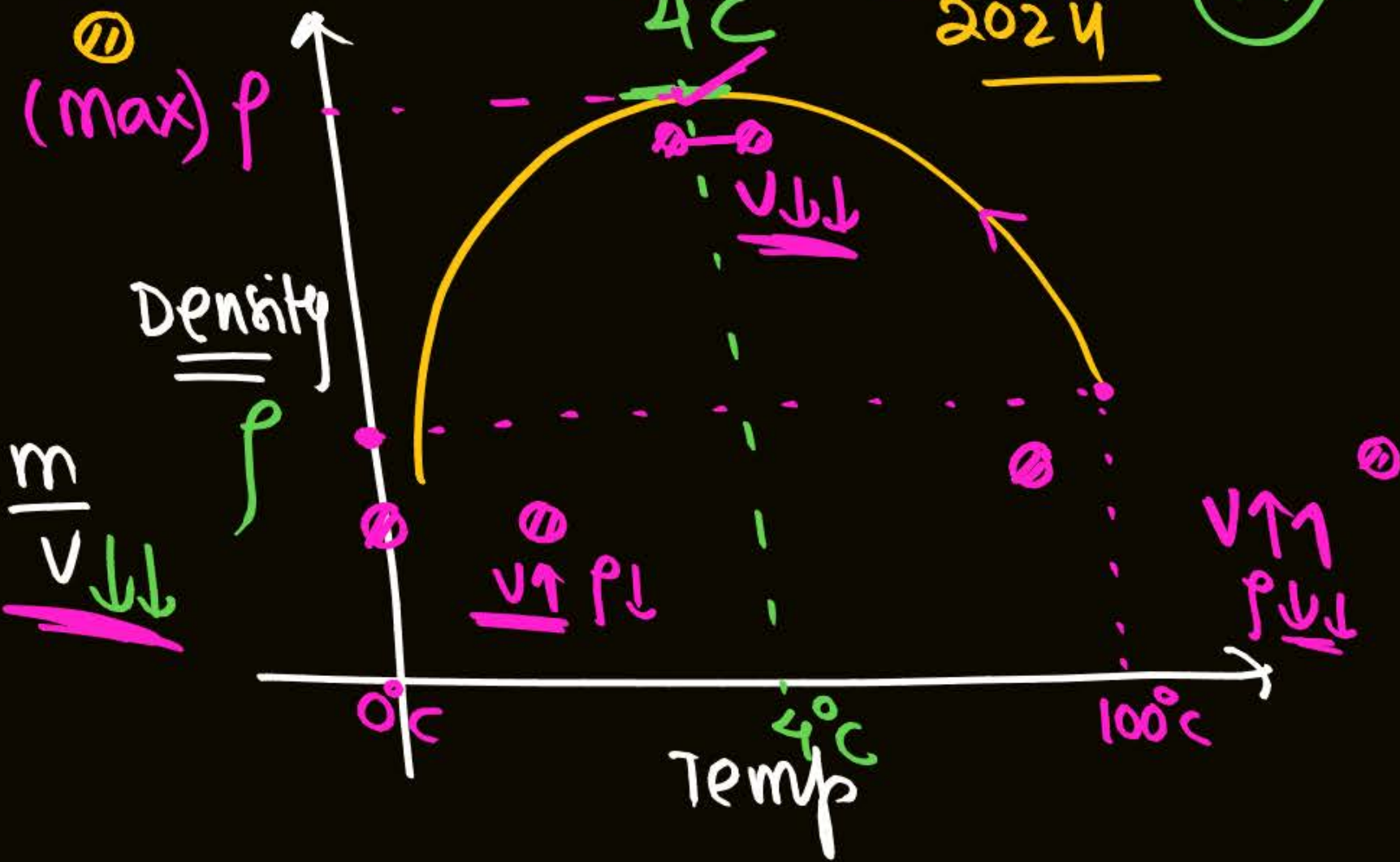
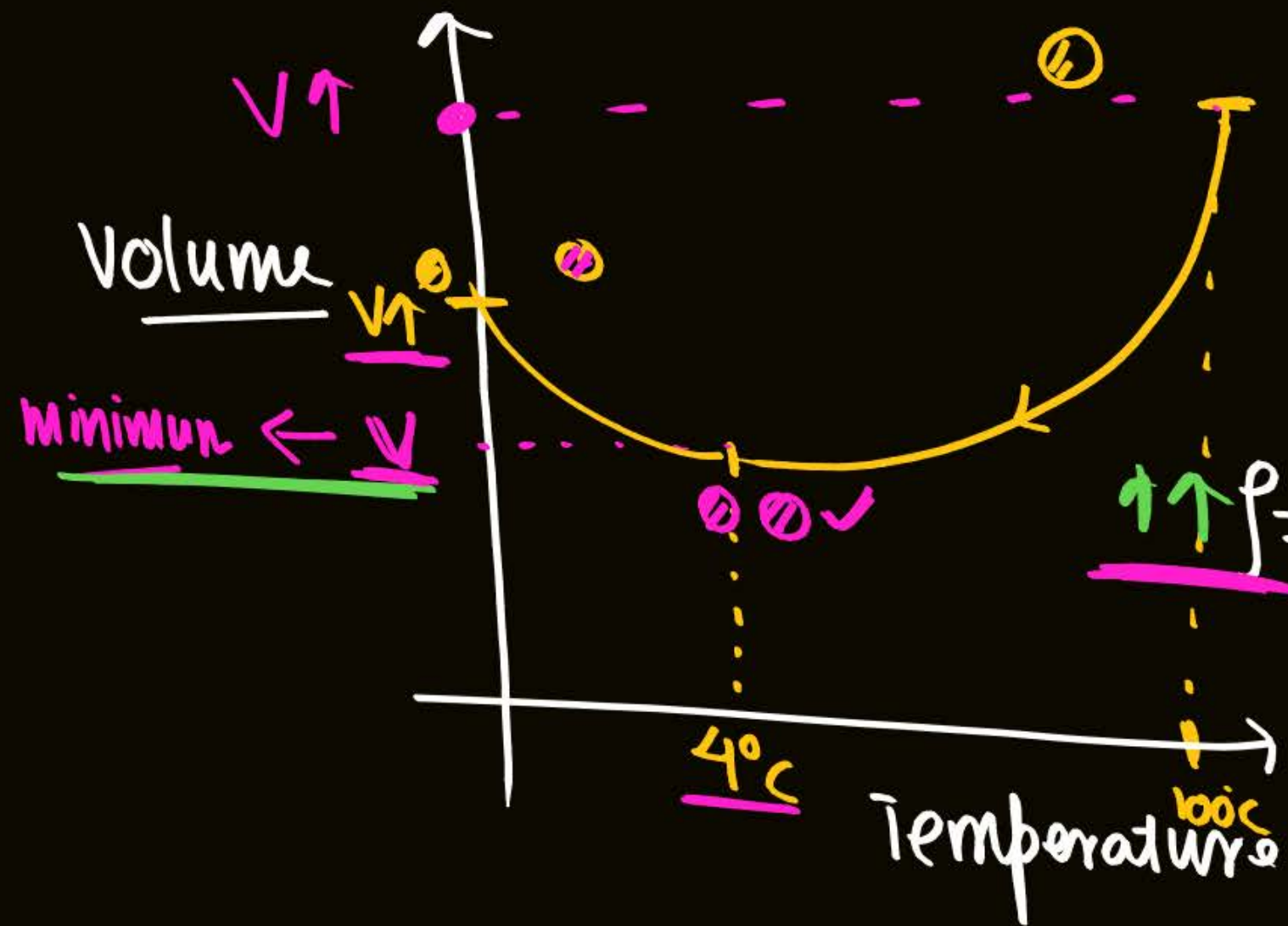
(a) minimum

☒ (b) maximum

(Graph Related to Abnormal behaviour of water)

Neet
2024

14



SUMMARY



① Temp $\uparrow$ $V \downarrow$ $\rho \uparrow \uparrow \uparrow$

$$\rho = \rho_0 (1 - \gamma \Delta T)$$

$$\begin{aligned} \text{② } \gamma_s &= \gamma \times \frac{\Delta \rho}{\rho_0} \\ &= \gamma \times \alpha \Delta T \end{aligned}$$

③ $4^\circ\text{C} \rightarrow \left. \begin{array}{l} V_{\min} \\ \rho_{\max} \end{array} \right\} \text{water}$

HOMework

minimum volume.

Q Why water have Max. density at 4°C?



Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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